

Just so, if we are confident that there are a huge number of different universes, we should not take the fact that we are in a life-permitting one to be evidence for much of anything.

So the key question is: do we have good reason to think that the multiverse hypothesis is true?

A first point to note: it would be very surprising if this hypothesis were true. For, if it is, there are very many — perhaps infinitely many — other universes, each as real as ours, in which some near-duplicate of you exists. There is, for example, very likely one in which there exists some being with a qualitatively identical history to you who differs from you only in that she or he scratched his nose one second ago.

This does not show that the multiverse hypothesis is false; the universe might be strange, and science repeatedly shows us that it is. But it does suggest that the multiverse hypothesis is not one that we should believe without argument.

One might think that the very facts used in the fine-tuning argument can be used to support the multiverse hypothesis. For consider the following argument:

But, while this reasoning sounds plausible, consideration of parallel cases shows
that something has gone wrong.

A Bayesian argument for the multiverse

It is very, very improbable that our universe is the only one and, just by chance, the constants came to be set in such a way as to make life possible. But if there were many many universes, it would not be very improbable that one would be life supporting. So, the fact that our universe is life-supporting is strong evidence in favor of the multiverse hypothesis.

A Bayesian argument for many dice rollers

I am sitting in my office, and I pick up 12 dice and decide to roll them. I roll all sixes. Amazed, I think to myself: there must be lots of people rolling dice in Malloy Hall right now. After all, what are the odds that someone rolls 12 sixes in Malloy in the case where there is just one person rolling dice?

This would be terrible reasoning; the fact that I rolled all sixes, however improbable, is not evidence for the existence of many rollers. What has gone wrong?



fine-
tuning of the
universe

A solid red circle with a subtle drop shadow, centered on a white background. Inside the circle, the text "Bayes' theorem" is written in a white, sans-serif font, centered both horizontally and vertically.

Bayes'
theorem



the
argument



objections

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Enough about probability (for now). Let's turn now to some results from

follows.

contemporary physics which will be important to the argument which

A first question: what does the theory provide by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)



These 'dimensional constants' will be our focus.

energy density of empty space.

One such constant is the cosmological constant, which measures the

The cosmological constant is just a number which (as far as we know)

could have different values consistent with the laws of physics.

(like the cosmological constant) would fall in a certain range.

likely it is, given the laws of the nature, that the fundamental constants

One thing that the standard model of physics gives us is a measure of how

hard to see how life could have evolved. If there were no planets, it is hard

We can make certain plausible assumptions about what it would take for

to see how life could have evolved.

life to have evolved. For example, if there were nothing but hydrogen, it is

constant has a value which would permit the evolution of life.

Given assumptions such as these, we can look at what the standard model

of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement.

(There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

supporting value, given the laws of nature, is very low.

This is the claim that the cosmological constant having a life-

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

nature, is

The claim is that, according to current physics, the probability of the

complogical constant falling in a life-supporting range, given the laws of



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probability theory into an argument for the existence of God.

Let's now ask how to turn these facts about current physics and

in the turn, so we can consider two different hypotheses about the origins

Much as we considered two different hypotheses about the number of balls

of the universe.

created by an intelligent designer. Let the non-design hypothesis be the

Let the design hypothesis be the hypothesis that the universe

hypothesis that it was not.

Let e be the fact that the cosmological constant has a

value in the life-permitting range.

Then what do we know about relevant probabilities?



$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$Pr(e|\text{design}) \equiv 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

With this information in hand, figuring out the probability of the non-

numbers.

design hypothesis given evidence is that just plugging in the



$$Pr(\text{non-design}) = 0.5$$



$$Pr(\text{design}) = 0.5$$



$$Pr(e) = 0.25$$



$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now

PowerBar!

It is difficult to think about numbers as large as the denominator of this fraction. But



consider the odds of winning Powerball one trillion times in a row. Call that a “super

Now consider the odds of winning a super Powerball one trillion times in a row. Call

that a 'super duper Powerball.'

Now consider the odds of winning a super Powerball one trillion times in a row.

The odds of this happening are about $1/10^{44}$ — so much, much higher than the

odd of the universe being life-permitting by chance.

This means that if you simply take the physics at face value, and begin by assigning a

the non-design hypothesis being true are vastly lower than the chances of winning a

probability of 0.5 to the non-design hypothesis, you should think that the chances of

superduperpowerball a trillion times in a row.

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1.

minus the very small number we're just discussing:



$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.



fine-
tuning of the
universe

A solid red circle serves as the background for the text.

Bayes'
theorem



the
argument



objections



$$3. P(\text{design}) = P(\text{non-design}) = 0.5.$$



2



P

$$C.P(God\ exists)\approx 1.(7,8)$$

6. *Life exists.*

independent designer, then God exists.

$$7. P(\text{design}) \approx 1. \quad (5, 6)$$

5



P



• (1, 2, 3, 4)

4. Bayes', theorem.

8. If the universe was created by an

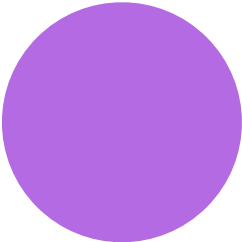
$$(life/death) \equiv 0.5$$

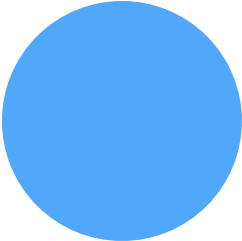
$$(1\text{ife} \mid \text{non-design}) = \frac{1}{10^{120}}$$

$$(\text{design} / \text{life}) \approx 1$$

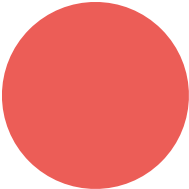






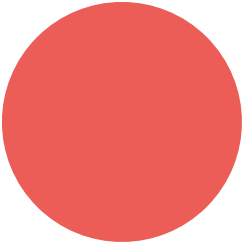








fine-
tuning of the
universe





cosmological constant falling in a life-supporting range, giving the laws of

nature, is

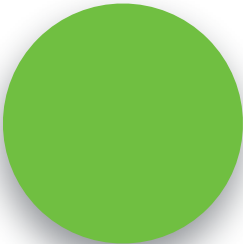
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